Dylan Thornburg

1. $V_{tn}=1$, $V_{tp}=-1V$ $\lambda=0$ $k'_n=40\,mA/V^2$ $k'_p=20\,uA/V^2$

a.

$I_2=0A,\ I_1=800uA$

$V_{DS}=10$ $V_{GS}=V2$

10 will be > $V2-V_T$ so it's in sat

saturation

$800uA=\frac{40uA}{2}\cdot 10(V2-V_T)^2$

$4=(V2-V_T)^2$

$2=V2-1$

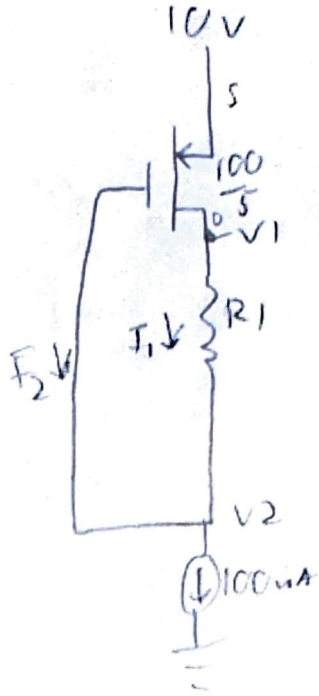
$V2=3V$

$4k\times 800uA=3.2V$

$V2+3.2=V1$

$V1=6.2V$

b.



$I_2=0A,\ I_1=100uA$

If $V_{DS}\le V_{GS}-V_T$ then in sat

$V1-10=V_{DS}$

$V_{GS}=V2-10$

$\frac{V1-V2}{5k}=100uA$ $V_{GS}-V_T=V2-10+1$

$V1=V2+.5$

$V1-10\le V2-10+1$

$V2-9.5\le V2-9$

$-9.5\le -9$ in sat

$100uA=20\cdot\frac{20uA}{V^2}(V2-9)^2$

$\frac{1}{4}=(V2-9)^2$

$\frac{1}{2}=(V2-9)$

$V2=9.5$ or 8.5

then $V1=10$ or 9

V1 can't be 10 so

$V2=8.5V$

$V2=8.5V,\ V1=9V$

ii If R1 is changed to 50Ω

$\frac{V1-V2}{50}=100uA$

$V1=V2+.005$

$V2-9.995\le V2-9$

true so still in sat.

iii If R1 is changed to 50kΩ

$\frac{V1-V2}{5k}=100uA$

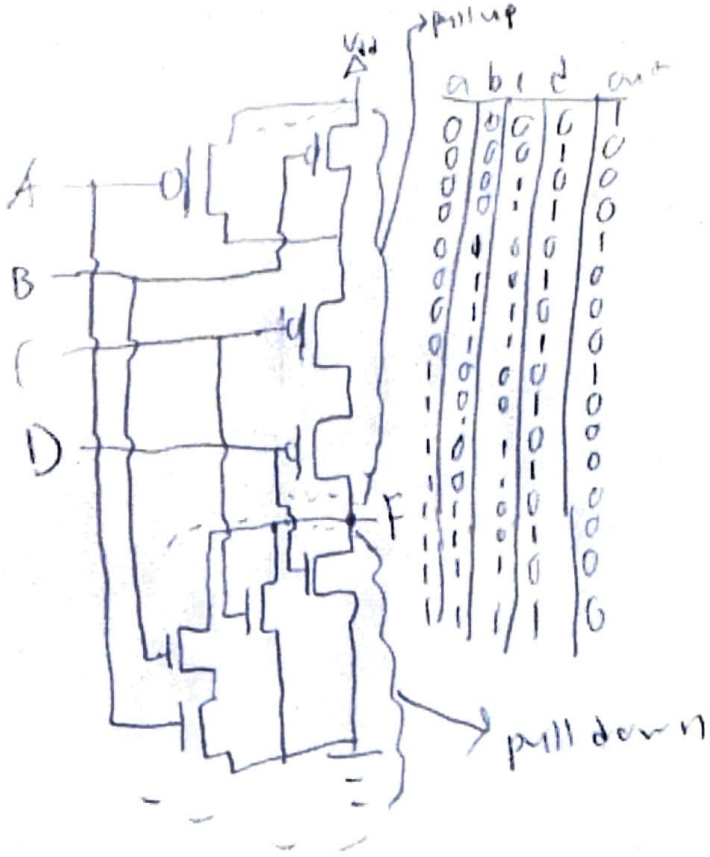
$V1-V2=5$

$V1=V2+5$

$V2-5\le V2-9$ false

in active now

2.



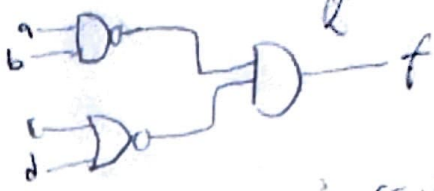
a	b	c	d	out
0	0	0	0	1
0	0	0	1	0
0	0	1	0	0
0	0	1	1	0
0	1	0	0	1
0	1	0	1	0
0	1	1	0	0
0	1	1	1	0
1	0	0	0	1
1	0	0	1	0
1	0	1	0	0
1	0	1	1	0
1	1	0	0	1
1	1	0	1	0
1	1	1	0	0
1	1	1	1	0

a or b has to be zero and c and d have to be zero

$(\bar{a}+\bar{b})(\bar{c}\cdot\bar{d})$

$(\overline{ab})(\overline{c+d})$

this circuit does this



$(\bar{a}+\bar{b})(\bar{c}\cdot\bar{d})$

$\bar{a}\bar{c}\bar{d}+\bar{b}\bar{c}\bar{d}$

Nands a and b, nors c and d, then ands the result

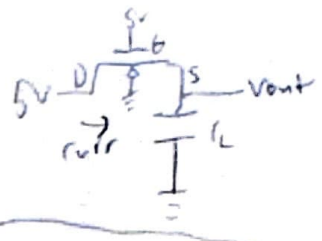
3.

$V_{TN}=1V$, $V_{TP}=-1V$ $\lambda=0$ $k'_n=\frac{20mA}{V^2}$ $k'_p=\frac{10mA}{V^2}$

a.

$V_f=0V$

$V_{in}=5V$ NMOS



$V_{out}=4V$

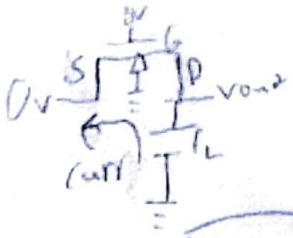
initial region: saturation

final region: cutoff

resistance $=\infty\Omega$ (open)

b. $V_f=5V$

$V_{in}=0V$



$V_{final}=1V$

initial region: cutoff

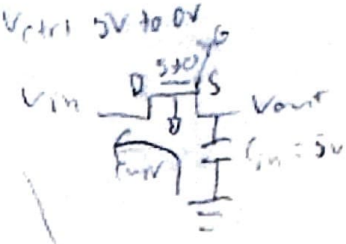
final region: triode region

resistance: 0Ω

c.

$V_f=5V$ $V_{in}=0V$ $V_{inv}=5V$

V_{ctrl} 5V to 0V



$V_{TP}=-1$ but source D flip so

$V_{finalout}=1V$

initial region: sat.

final region: cutoff

V_{gsout}

resistance $=\infty\Omega$ (open)

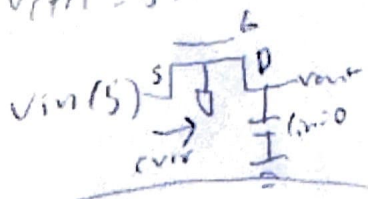
[redacted]

[redacted]

3 cont

d. $V_{ctrl} = 0V$ $V_{in} = 5V$

$V_{ctrl} = 5 \to 0$



Vout final = 4V

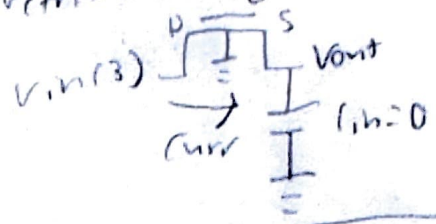
initial region = saturation

final region = triode

resistance = 0Ω

e. $V_{ctrl} = 0V$ $V_{in} = 3V$

$V_{ctrl} = 0 \to 5$



3-0 3-3=0

vs 5-0

3 ≤ 5-1 ✓ final

3-3 vs 5-3

0 ≤ 2-1 ✓

Vout final = 3V

initial region = triode

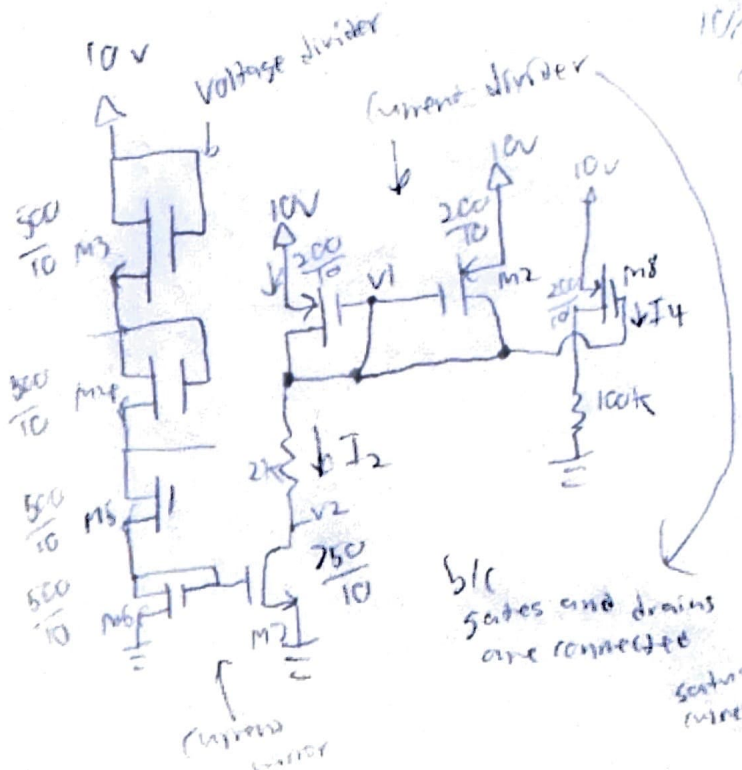
final region = triode

resistance = 0Ω

4. $V_{TP} = -1.3V$ $k_n' = 25\mu A/V^2$ $k_p' = 15\mu A/V^2$

$\lambda = 0$

$V_{TN} = 1.3V$



10/4 = 2.5V all of the voltage divider NMOS are in sat b/c shown said so. but also

10-2.5 ≤ 10-7.5-1.3 false so in sat

$V_{GS} = 2.5V$

$I_3 = \frac{25\mu A}{2} \cdot \frac{500}{10}(2.5-1.3)^2 = .9mA$

$I_2 = \frac{\frac{250}{10}}{\frac{500}{10}}(.9mA) = 1.35mA$

$I_1 = \frac{1}{2}I_2 = .675mA$

b/c gates and drains are connected

saturation b/c current dividers are saturated

$.675mA = \frac{15\mu A}{V^2}\frac{200}{10}(V_{gs}+1.5)^2 \cdot \frac{1}{2}$

$20 = (V_{gs}+1.5)^2$

$\sqrt{20} = V_{gs}+1.5$

$V_{gs} = -3.62$

$V_1 = 2.5 - 10 = -7.5$

Next.

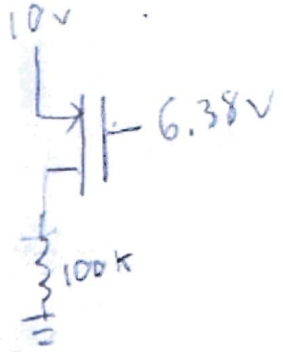
$$\frac{6.38 - V2}{2k} = 1.35\,mA$$

$V2 = 3.68V$

a. $V1 = 6.38V$
$V2 = 3.68V$
$V3 = 2.5V$ unless you measure to ground across the other 2 transistors in which it would be 2.5V

$I1 = .675\,mA$
$I2 = 1.35\,mA$
$I3 = .9\,mA$
$I4 =$ no curr, 0A

b.



$6.38 - 10 = V_{GS} = -3.62V$

$-1.3 > -3.62$ so not cutoff

$-10 < -3.62 - 1.5$

$-10 \leq -2.12$

$V_{DS} \leq V_{GS} - V_{TP}$

so we are in saturation mode